

Mock Exam (L3 - Ms. Maria Samad)

CS 101 Algorithmic Problem Solving

Fall 2023

Name(s): _____

HU ID (e.g., xy01042): _____

1. SALE! SALE! SALE!

Your friend is drawn to a Crazy Sale at a toy shop run by a retired CS professor. Old age has caught up with her and she mistakes her customers to be capable CS students. At the sale, she applies a discount on the total bill as follows.

- If 1 item is bought, the discount is 10%
- For every subsequent item less than 5 items, there is an additional discount of 2%
- If 5 items are bought, the discount is 20%
- For every subsequent item less than 10 items, there is an additional discount of 5%
- If 10 items are bought, the discount is 50%
- For every subsequent item, there is an additional discount of 10%
- The total discount does not exceed 70% regardless of the number of items bought.

Your friend's head spins at this but fortunately for them, you are a capable CS student. You agree to write your friend the exact code that will tell them the discount rate corresponding to the number of items bought.

Constraints

- `num_items` $\in \mathbb{N}$
- `num_items` ≥ 0

Interaction

The input consists of an integer `num_items`, which represents the number of items being bought.

The output contains the rate applied on the `num_items`.

Sample

Input	Output
0	0
12	70

In the first case, no items are bought, hence no discount applied.

In the second case, The discount rate is 50% for 10 items and another 20% for the additional 2 items, so a total of 70%

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4	
5	
17	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

2. Money Changer

PakStreet Financial Services allows you to receive your money in any denominations. However, all the customers have started taking money in highest denominations, which has left them with uneven distribution in the currency notes.

The manager then suggested to divide the amount in all possible denominations starting from the highest and then give it to the customers. The employees, however, are having great difficulty in calculating that each time with so many customers, so they have hired a Computer Scientist to write them a code that will tell them the exact number of currency denominations that must be given out for the amount payable to them.

Constraints

- amount $\in \mathbb{N}$
- amount ≥ 1

Interaction

The input contains the total amount that the customer is supposed to receive.

The output gives the total number of currency bills and their quantity for the given amount on separate lines

Sample

Input	Output
10200	5000 : 2
	1000 : 0
	500 : 0
	100 : 2
	50 : 0
	20 : 0
	10 : 0
	5 : 0
	2 : 0
	1 : 0
2769	5000 : 0
	1000 : 2
	500 : 1
	100 : 2
	50 : 1
	20 : 0
	10 : 1
	5 : 1
	2 : 2
	1 : 0

In the first case, the total amount is Rs.10200, so starting from highest denominations, the customer will get 2 Rs.5000 bills. For the remaining amount, i.e. Rs.200, the highest denomination is Rs.100, so they get 2 Rs.100 bills.

Similarly, in the second case, the first highest denomination can be of Rs.1000, so the customer gets 2 of them, 1 of Rs.500 bill, 2 of Rs.100 bills, 1 Rs.50, 1 of Rs.10, 1 of Rs.5, and 2 of Rs.2 to make the total of Rs.2769

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
7813	
570	
9	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

3. Generate Patterns with Nested Iterations

Write an iterative function named `pattern` to generate the given pattern for a given parameter, n

Constraints

- $n \in \mathbb{N}$
- $n > 0$

Interaction

The input contains a single integer, n and using that n value, the patterns can be printed.

The output is the desired number pattern printed on screen.

Sample

Input	Output
2	1 2 1
6	1 2 1 4 2 1 8 4 2 1 16 8 4 2 1 32 16 8 4 2 1

In the first case, the input is 2, so it prints the desired pattern in two lines.

In the second case, the input is 6, so the pattern is printed on 6 lines.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
3	
10	
7	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

4. Candy Bag

Abeeha has two group of friends she hangs out with. One group has 5 people while the other group has 8 people. She goes to buy candies for her friends. She cannot buy candies individually, only bags of candies are available. Given the quantity of candies in a bag, she needs to figure out which group will have evenly distributed candies. She basically needs to figure out which group will have the lesser number of remaining candies after the candies have been evenly distributed.

Write a function called *candy_bag* that takes N - the number of candies in a bag, and returns the group for which it would be more suitable (better even distribution).

Group with 5 people is group 1 and the other is group 2. If both groups are suitable then return 0.

Don't forget to call the function and print the outputs, as desired.

Constraints

- $10 \leq N \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of N .

The output must contain a single number denoting whether the candy bag is suitable for group 1 or group 2, or either one.

Sample

Input	Output
100	1
73	2

In the first case, if 100 candies are distributed among 5 people, there will be no remainder however, if they are distributed among 8 people then Abeeha will be left with 4 candies, hence, group 1 is preferable.

In the second case, if 73 candies are distributed among 5 people Abeeha will be left with 3 candies. However, with second group she will only be left with 1 candy. Therefore, group 2 is more suitable.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
40	
154	
101	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

TIME TO DO SOME DEBUGGING!**5. Dark Light**

You have a special torch, this torch has 4 levels numbered from 1 to 4 and 2 states (On and Off), Levels 1,2 and 3 correspond to the On state while level 4 corresponds to the Off state.

The levels can be changed as follows:

- Level 1 changes to level 2
- Level 2 changes to level 3
- Level 3 changes to level 4
- Level 4 changes to level 1

Given the initial state as K and the number of changes made in the levels as N , write a function state which takes these parameters and returns the final state of the torch. If the final state cannot be determined, return 'Ambiguous' instead.

Constraints

- $1 \leq N \leq 10^9$
- $0 \leq K \leq 1$

Interaction

The input comprises of two space separated integers denoting the value of N, K .

The output must contain a string either being Off, On or Ambiguous depending on the criteria told above.

Sample

Input	Output
4 0	Off
3 1	Ambiguous

In the first case, $(N, K) = (4, 0)$, initially the torch is on Off state. Since only level 4 corresponds to the off state, the torch is at level 4 initially. After 4 changes, the final level would be 4 and that corresponds to the Off state.

In the second case, $(N, K) = (3, 1)$, initially the torch is on On state, this can correspond to any of the first 3 levels, thus we cannot determine the final level after 3 changes, hence the final state is Ambiguous.

Proposed Solution

```
1 def state(n,k):
2     if k == 0:
3         if (n % 4 == 0):
4             print("Off")
5     else:
6         if (n % 4 != 0):
7             print("Ambiguous")
8
9     print(state(k,n))
10
```

Dry Run

Using any one of the inputs provided above, dry run your pseudocode in the space below.

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Rough Work